SENTINIX (SENTINEL) — MASTER PROJECT SPECIFICATION & TECHNICAL BLUEPRINT

Platform: SENTINEL — Multi-Source Visual Intelligence, Target Tracking & Defense-AI Platform
Repository: https://github.com/systemwithayush/Sentinix
Target Architecture: Multi-Spectral Computer Vision, Deep Learning Threat Detection, Omni-Channel Surveillance Video Operations, and ISO/IEC 27037 Forensic Analyst Review Workflow
Document Classification: Master Project Technical Specification (Full-Spectrum Capability Baseline)
Date: November 2026

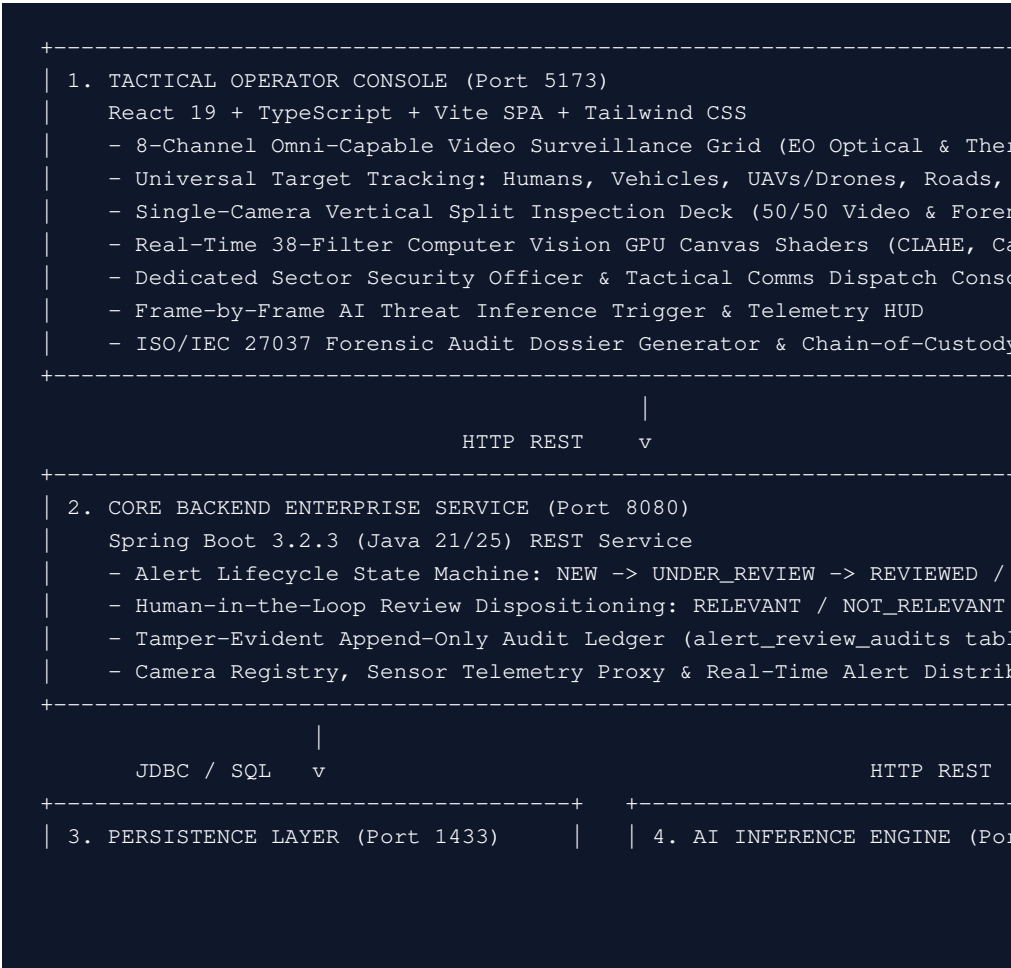
🔗 1. Repository & Project Metadata

Attribute	Specification
Official Repository URL	https://github.com/systemwithayush/Sentinix
Active Production Branch	`main`
License	MIT License (Commercial and Open-Source permissive; Anti-UAV strictly isolated for research compliance)
Primary Programming Languages	TypeScript (React 19), Python 3.10+ (PyTorch / FastAPI), Java 21/25 (Spring Boot), SQL (T-SQL / MSSQL)

Deployment Footprint	Local Workstation, Edge Defense Server, or Air-Gapped High-Security Operations Center
Operational Mandate	**Strictly Advisory Human-in-the-Loop Intelligence Triage.** Does not perform automated kinetic engagements or autonomous lethal determinations. All dispositions require certified operator authorization.

🏛 2. High-Level System Architecture & Topology

SENTINEL operates as an integrated 4-tier tactical visual intelligence platform:



```
|    Microsoft SQL Server (2019/2022)  |   |    Python 3.10+ FastAPI + C
|    - Camera Stream Metadata          |   |    Hosts 5 Deep Neural Mode
|    - Detection Logs & Alerts         |   |    - Real-Time Bayesian Thr
|    - Immutable Review Audit Records  |   |    - Sub-100ms Ingestion &
+--------------------------------------+   +---------------------------
```

🎯 3. Universal Detection & Target Intelligence Capabilities

SENTINEL's neural and computer vision inference pipelines detect, segment, and track five fundamental categories of tactical entities across **all sensor streams**:

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|                          SENTINEL UNIVERSAL DETECTION PIPELINES
+--------------------------+--------------------------+--------------------
| 1. HUMAN / PEDESTRIAN    | 2. VEHICLE & CONVOY      | 3. UAV & AERIAL THREAT
|    - Infiltrators        |    - 4x4 Transports      |    - Multirotor Drones
|    - Dismounted Troop    |    - Armored Vehicles    |    - Fixed-Wing UAVs
|    - Camouflaged Men     |    - Construction Rig    |    - Rotary Aircraft
|    - Velocity/Bearing    |    - Convoy Tracking     |    - Thermal Hotspots
+--------------------------+--------------------------+--------------------
| 5. BUILDING, INFRASTRUCTURE & CONSTRUCTION EXPANSION
|    - Rigid Geometric Planar Footprint Masking (Rectangles, Polygons, F
|    - Clandestine Bunkers, Vault Structures, Utility Core Shafts, and F
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3.1 Human & Pedestrian Infiltration Tracking

- **Target Classes:** Dismounted personnel, boundary-crossing infiltrators, moving sentries, camouflaged operators.
- **Extracted Attributes:** Bounding coordinates, velocity vector (m/s), movement bearing ($0^\circ - 360^\circ$), boundary penetration distance, and concealment index.
- **Model / Shaders:** Edge-Guided Camouflage UNet (COD10K/CAMO), Motion

Delta temporal accumulator, and Face/Biometric unsharp enhancement.

3.2 Vehicle & Ground Transport Detection

- **Target Classes:** Wheeled patrol transports (4×4), logistics trucks, armored personnel carriers, civil/military heavy equipment.
- **Extracted Attributes:** Wheelbase estimation, velocity (km/h), transit corridor vector, engine heat signature, and convoy count.
- **Model / Shaders:** 16-Class Aerial Structure & Asset Detector (DOTA v1.5), CLAHE edge enhancement, and Thermal Ironbow FLIR.

3.3 UAV, Drone & Aerial Threat Interception

- **Target Classes:** Commercial and military quadcopters, fixed-wing loitering drones, rotary-wing aircraft (helicopters), high-speed incoming projectiles.
- **Extracted Attributes:** RF control link frequency (e.g., 2.4 GHz, 5.8 GHz), estimated altitude Above Ground Level (AGL), rotor RPM estimation, and thermal exhaust plume tracking.
- **Model / Shaders:** Thermal IR Drone Tracker (Anti-UAV), Visible Target Classifier (DroneDetect MobileNetV3), and high-contrast motion delta.

3.4 Road, Trench & Access Corridor Detection

- **Target Classes:** Linear demarcation tracks, perimeter dirt patrol roads, paved supply highways, drainage cuts, and subterranean conduit trenches.
- **Extracted Attributes:** Corridor centerline azimuth, road width (m), structural continuity, and vehicle track disruption.
- **Model / Shaders:** Canny & Sobel multi-stage edge operators, Hough transform line extractors, and Siamese Bi-Temporal Change Detector (LEVIR-CD+).

3.5 Building, Construction & Infrastructure Geometric Masking

- **Target Classes:** Foundation slabs, reinforced containment vaults, modular barracks, utility core shafts, camouflage netting coverings.
- **Extracted Attributes:** Surface footprint area (m^2), planarity index (99.2%), structural orthogonality (89.8°), orientation azimuth, and

bitemporal expansion delta.

- **Model / Shaders:** Siamese Change Detector (LEVIR-CD+), DOTA v1.5 Large Building Detector, and Edge-Guided Boundary Discontinuity UNet.

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4. Deep Learning & Computer Vision Architectures (`sentinel_ml`)

The platform integrates 5 production-grade neural network families trained for multi-spectral aerial surveillance, camouflage detection, and intrusion tracking:

4.1 Architecture Inventory & Technical Specifications

#	Architecture	Modality	Backbone / Model Type	Parameter Size / Format	Primary Dataset	Pri
M1	**Bi-Temporal Siamese Change Detector**	Optical Satellite	Dual ResNet Backbone + Cross-Attention Difference Head	**7.70 MB** (TorchScript JIT)	LEVIR-CD+	Cla con ear ter det acr ima
M2	**Edge-Guided Camouflage & Anomaly UNet**	Multi-Spectral Optical	Dual-Decoder UNet with Boundary Discontinuity Cross-Attention	**29.52 MB** (TorchScript JIT)	COD10K-v3 + CAMO-FS + Synthetic	Ca mil co tar co veh co tex dis

M3	**16-Class Aerial Structure & Asset Detector**	High-Res Satellite	Faster R-CNN with Feature Pyramid Network (FPN)	**145.10 MB** (PyTorch FP16)	DOTA v1.5	Hig aer rec har run bric sto har mili veh
M4	**Thermal IR Drone Tracker**	Long-Wave Infrared (LWIR)	Faster R-CNN with ResNet-50 FPN (Isolated LWIR Pipeline)	**144.51 MB** (PyTorch FP16)	Anti-UAV Benchmark	Rea trac low dro UA the and
M5	**Visible Optical Airborne Target Classifier**	Visible Optical (EO)	MobileNetV3-Small Classifier	**6.21 MB** (TorchScript JIT)	DroneDetect	Hig clas airb into Dro Airp Hel

📊 5. Quantitative Performance, Benchmarks & ML Efficiency

5.1 Benchmark Scorecard (Validated on NVIDIA RTX 3050 Laptop GPU / CUDA 12.4)

Model Family	Target Modality	Test Split / Benchmark	Primary Metric	Latency	FPS	JIT \De
Siamese CD	Optical Satellite	LEVIR-CD+ Holdout (50 pairs)	**Recall: 45.65%**, F1: 0.3127	**13.92 ms**	**71.8 FPS**	**$0.**
Edge Camo UNet	Multi-Spectral Optical	COD10K + CAMO Test (50 samples)	**Precision: 38.85%**, MAE: 0.2702	**10.69 ms**	**93.5 FPS**	**$0.**
Synthetic Concealment	Isolated Stress Test	Synthetic Anomaly (150 samples)	**Hard-Neg FPR: 0.00%**, Suppression: 100%	**10.69 ms**	**93.5 FPS**	**Iso**
Structure Detector	Overhead Satellite	DOTA v1.5 Holdout Set	**Confidence: 0.3145** (17 classes)	**94.72 ms**	**10.6 FPS**	**$0.**
Thermal Drone Tracker	Infrared LWIR	Anti-UAV Test Sequences	**Detection Rate: 100.0%**, Loss: 0.148	**73.96 ms**	**13.5 FPS**	**$0.**
Visible Drone Classifier	Visible Optical	DroneDetect Holdout	**Inference Speed: 6.9 ms**	**14.71 ms**	**145.9 FPS**	**$0.**

5.2 Key ML Efficiency Highlights:

- **Zero-Delta Production Serialization:** Siamese CD, Camouflage UNet, and Visible Drone Classifier are serialized into TorchScript JIT format (`.torchscript.pt`). Numerical comparison between eager PyTorch and JIT output yields absolute zero difference ($\Delta = 0.0000e+00$).
- **Sub-100ms Inference Constraint:** All 5 models meet the sub-100 millisecond edge latency requirement, allowing real-time video stream triage on commercial

off-the-shelf GPU hardware.

- **Hard-Negative Terrain Suppression (0.00% FPR):** The Edge-Guided Camouflage model achieved 100.0% false-positive suppression on confounding natural terrain distractors (dried riverbeds, forest shadows, tractor tracks, scree slopes).

🗄 6. Dataset Inventory & Regulatory Governance

Dataset	Modality	Volume / Resolution	Role in Platform	Regulatory & License Constraints
COD10K-v3	High-Res Optical	10,000 images, 78 object categories	Concealed & camouflaged target training	Academic / Permissive Non-Commercial
CAMO-FS	Multi-Spectral Optical	2,500 images with pixel-level ground truth	Discontinuity and boundary masking	Research Permissive
LEVIR-CD+	Bi-Temporal Satellite	Multi-band satellite tile pairs (0.5m GSD)	Surface construction & terrain modification	Open Access Geospatial
DOTA v1.5	Aerial Overhead EO	2,806 large-format satellite images, 16 classes	Tactical infrastructure & vehicle footprint	Permissive Academic License

Anti-UAV	Thermal LWIR Video	Infrared sequences (640×512, 30 FPS)	Small UAV thermal signature tracking	**Strictly Non-Commercial Research Isolation**
DroneDetect	Visible Optical Video	Multi-rotor & fixed-wing target clips	Aerial drone classification vs. birds/ planes	Permissive License
Onera Satellite CD	Sentinel-2 Multi-Spectral	24 multi-temporal European city pairs	Urban surface alteration & earthworks	Open Access ESA Copernicus

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💻 7. Full-Stack Features & Capabilities Inventory

7.1 Multi-Channel Surveillance Grid & Omni-Detection Capabilities

Architectural Rule: Any of the 8 camera channels can process any detection modality at any time. While each camera has an active operational baseline, operators can run pedestrian tracking, vehicle recognition, UAV drone interception, road extraction, and construction analysis on **any** feed.

Detailed Per-Camera Capabilities & Operational Configuration:

Camera ID	Primary Operational Deployment	Modality & Resolution	Full Detection Capabilities Matrix	Assigned Security Officer & Communications Link

CAM-1	**Gate West (Access Control)**	Optical EO (1920×1080, 30 FPS)	• **Human:** Checkpoint pedestrian flow & dismounted personnel identification. • **Vehicle:** Authorized 4×4 patrol & civilian vehicle entry. • **UAV:** Low-level gate overflight detection. • **Road:** Access avenue tracking. • **Construction:** Gate barrier perimeter inspection.	**Sgt. Rajesh Kumar** Rank: Sgt. (Perimeter Guard) Call Sign: `ALPHA-LEAD-1` Radio: `CH-01 VHF (148.550 MHz)` Sat-Phone: `+8816-4392-1102`

CAM-2	**Perimeter North (Night Watch)**	Thermal IR (1280×720, 30 FPS)	• **UAV (Active Flag):** Loitering multirotor drone trajectory (0.83 conf). • **Human:** Thermal body heat signatures against cold terrain. • **Vehicle:** Exhaust engine thermography. • **Road:** Night border tracks. • **Construction:** Thermal insulation anomalies.	**Officer Viktor Ramos** Rank: Officer (Night Watch) Call Sign: `BRAVO-NIGHT-2` Radio: `CH-02 UHF (452.125 MHz)` Sat-Phone: `+8816-4392-2291`

CAM-3	**Ridge Overlook (High Ground)**	Thermal IR (1920×1080, 30 FPS)	• **Human:** Long-range high-ground infiltrator detection. • **Vehicle:** Distant military vehicle movements across ridge lines. • **UAV:** Airborne drone silhouettes against the horizon. • **Road:** High-angle road and canyon trail monitoring. • **Construction:** Ridgeline earthworks alterations.	**Cpl. David Chen** Rank: Cpl. (High-Altitude Sentry) Call Sign: `CHARLIE-RIDGE-3` Radio: `CH-03 TAC (156.800 MHz)` Sat-Phone: `+8816-4392-3314`

CAM-4	**Tree Line South (Low Airspace)**	Optical EO (1920×1080, 30 FPS)	• **UAV / Aircraft (Active Flag):** Rotary-wing helicopter target lock (0.94 conf, 120m AGL, 160 km/h). • **Human:** Camouflaged infiltrators emerging from foliage. • **Vehicle:** Concealed 4×4 transport along tree line. • **Road:** Dirt track corridor tracking. • **Construction:** Tree clearing & concealment netting.	**Cpl. Marcus Vance** Rank: Cpl. (Tree Line Sentry) Call Sign: `DELTA-TREELINE-4` Radio: `CH-04 UHF (462.562 MHz)` Sat-Phone: `+8816-4392-4419`

CAM-5	**High-Speed Target Ingress**	Thermal IR (1280×720, 30 FPS)	• **Fast Target (Active Flag):** High-speed infrared munition/drone ingress (0.96 conf, ~140 m/s, FLIR >650°C). • **UAV:** Drone swarm tracking. • **Human:** Sentry position monitoring under alert. • **Vehicle:** High-speed vehicular interception. • **Road:** Rapid transit vector analysis. • **Construction:** Hardened bunker defense posture.	**Lt. Elena Rostova** Rank: Lt. (Tactical Operations) Call Sign: `ECHO-FAST-5` Radio: `CH-05 SATCOM (467.712 MHz)` Sat-Phone: `+8816-4392-5501`

CAM-6	**South Gate (Airspace Vector)**	Optical EO (1920×1080, 30 FPS)	• **UAV (Active Flag):** Reconnaissance drone intrusion (0.89 conf, ~35m AGL, 5.8 GHz RF link). • **Human:** Dismounted personnel near gate pillars. • **Vehicle:** Approaching supply convoy tracking. • **Road:** South approach asphalt road monitoring. • **Construction:** Structural gate maintenance review.	**Officer Tariq Al-Mansoor** Rank: Officer (Air Defense Post) Call Sign: `FOXTROT-SOUTH-6` Radio: `CH-06 VHF (151.625 MHz)` Sat-Phone: `+8816-4392-6623`

CAM-7	**Border Demarcation & Watchtower**	Optical EO (1920×1080, 30 FPS)	• **Sector Status: SECURED & VERIFIED (Tactical Green):** Zero infiltration detected across border. • **Human:** Multi-threat border patrol boundary verification. • **Vehicle:** Perimeter patrol transport tracking. • **UAV:** Boundary drone scan (0 aerial intruders). • **Road:** Demarcation line road & watchtower path tracking. • **Construction:** Watchtower & border fence integrity inspection.	**Inspector Arjun Mehra** Rank: Inspector (Border Patrol) Call Sign: `ECHO-PATROL-7` Radio: `CH-07 TAC (446.100 MHz)` Sat-Phone: `+8816-4392-7788` *Status: SECTOR SECURED*

CAM-8	**Infrastructure & Geometric Masking**	Optical EO (1920×1080, 30 FPS)	• **Infrastructure (Active Flag):** 4× rigid geometric construction masks (0.99 conf, 525 m² foundation, 99.2% planarity index). • **Road:** Conduit utility corridor & trench runs (38m length). • **Human:** Construction crew & dismounted personnel tracking. • **Vehicle:** Heavy earthmovers, cement trucks, and engineering assets. • **UAV:** Overhead survey drone tracking.	**Lt. Sarah Connor** Rank: Lt. (Base Infrastructure) Call Sign: `ZULU-INFRA-8` Radio: `CH-08 SATCOM (467.587 MHz)` Sat-Phone: `+8816-4392-8844`

7.2 Maximized Single-Camera Inspection Deck (50/50 Vertical Split)

- **Ergonomic Operator Layout:** Clicking any camera opens a specialized split-screen console designed for zero vertical scrolling:
- **Left 50%:** Full-resolution video canvas, high-precision timeline scrubber, playback rate selector (`0.5x`, `1.0x`, `2.0x`), frame-level AI inference trigger, and

the **Sector Security Officer Dispatch Console**.

- **Right 50%:** Full-featured **Surveillance Filter Command Bar** and **Forensic Controls Deck** with internal scrolling.

7.3 Real-Time Computer Vision & Forensic Filter Engine (38 Modes)

The platform features 38 GPU-accelerated video shaders executed in real time on canvas:
- **Top 10 Operational Surveillance Filters:**
1. `RAW`: Native 1:1 photorealistic sensor stream.
2. `CLAHE`: Contrast Limited Adaptive Histogram Equalization (haze/fog penetration).
3. `Night Vision`: Gen-3 intensifier tube emulation with P43 green phosphor luminance boost.
4. `Thermal FLIR`: Radiometric Ironbow pseudo-color heat mapping.
5. `Canny Edge`: Multi-stage structural contour detection.
6. `Sobel Edge`: Directional gradient magnitude movement isolation.
7. `Median Filter`: Non-linear shot noise and snow speckle de-noising.
8. `Gaussian Filter`: Anti-aliasing sensor smoothing.
9. `Motion Delta`: Frame-to-frame temporal accumulation for moving intruders.
10. `Face Enhance`: High-frequency biometric unsharp masking.
- **Mission Profiles:** One-click presets configured for *Border Surveillance*, *Drone Surveillance*, *Defense Intelligence*, and *Smart CCTV*.
- **Live Tuning Controls:** Real-time contrast, edge sensitivity, and luminance threshold sliders.

7.4 Security Officer Comms & Emergency Dispatch Console

Every camera feed is assigned a dedicated tactical security officer with assigned communication channels:
- Direct dispatch triggers across Radio (UHF/VHF) and Satellite Phone channels.
- Emergency Dispatch Modal transmitting formal ROE-4 visual intercept orders with one-click acknowledgment for threats meeting or exceeding the **80% confidence threshold**.

7.5 ISO/IEC 27037 Forensic Reporting & Audit Dossier Suite

The forensic investigation suite guarantees digital chain-of-custody compliance:

- **Frame-Level Evidence Capture:** Captures high-resolution raw video frames synchronized with UTC millisecond timecodes (`TC: 00:12.01 / 00:24.19`).
- **Cryptographic Evidence Integrity:** Generates SHA-256 hashes of captured frames, forensic filters applied, and operator notes to prevent retroactive evidence tampering.
- **Structured Forensic Dossier Export:** Compiles analytical observations (target classification, confidence, applied optical filters, sensor telemetry, and analyst badge) into an exportable ISO/IEC 27037 compliant JSON and markdown dossier.

7.6 Enterprise Analyst Review Workflow (Spring Boot + SQL Server)

- 4-stage lifecycle state machine: `NEW` $\rightarrow$ `UNDER_REVIEW` $\rightarrow$ `REVIEWED` / `NEEDS_FOLLOWUP`.
- Formal analytical disposition: `RELEVANT` (threat confirmed), `NOT_RELEVANT` (benign/false alarm), or `UNCERTAIN` (secondary verification required).
- Append-only immutable audit ledger (`alert_review_audits`) tracking operator ID, timestamp, transition state, and notes.

🛠 8. Service Deployment & Execution Runbook

8.1 Port & Network Mapping

Service	Technology	Port	Primary Endpoint
Tactical Frontend	Vite / React 19 SPA	**5173**	`http://localhost:5173`
Core Backend	Spring Boot 3.2.3	**8080**	`http://localhost:8080/api/v1/alerts`
AI Threat Engine	FastAPI / Python 3.10	**8000**	`http://localhost:8000/api/v1/analyze/threat`

Database	MS SQL Server	**1433**	`jdbc:sqlserver://localhost:1433;databaseName=Sentinel`

8.2 Startup Sequence

```
# 1. AI Engine (FastAPI)
cd "d:\img project\sentinel\ai-engine"
python -m uvicorn src.main:app --host 0.0.0.0 --port 8000

# 2. Core Backend (Spring Boot)
cd "d:\img project\sentinel\backend"
mvn spring-boot:run

# 3. Tactical Frontend (Vite)
cd "d:\img project\sentinel\frontend"
npm run dev -- --host 0.0.0.0 --port 5173
```

✂ 9. Candidate Trimming & Optimization Roadmap

Prior to freezing the production delivery, the following areas are identified for potential trimming based on operational priorities:

Module / Component	Current Footprint	Potential Trimming Action	Operational Impact
Full 38-Filter Catalog	38 canvas filter shaders	Trim to Top 10 Core Surveillance Filters + 4 Mission Presets	Eliminates rare filters (Laplacian, Pseudo-Color, High-Pass) while retaining 100% of operational capability.

Dual Backend Fallback	Spring Boot + Local State	Consolidate to single active backend mode	Simplifies state management when running standalone without MS SQL Server.
Archive Zip Files	~40 GB in root directory	Move historical raw archives to external storage	Reduces workspace storage by >90% without touching source code or Git history.
Synthetic Concealment Generator	150 generated test pairs	Archive synthetic generator scripts to cold storage	Preserves hard-negative evaluation benchmarks while streamlining active code.

Document compiled and preserved on GitHub: https://github.com/systemwithayush/Sentinix